

Computational Concept of the Psyche

Anton Kolonin^{1,2,3} [0000-0003-4180-2870]

akolonin@gmail.com

Vladimir Krykov² [0009-0008-4930-4618]

¹ Novosibirsk State University, Pirogova 1, 630090, Novosibirsk, Russia, <https://nsu.ru>

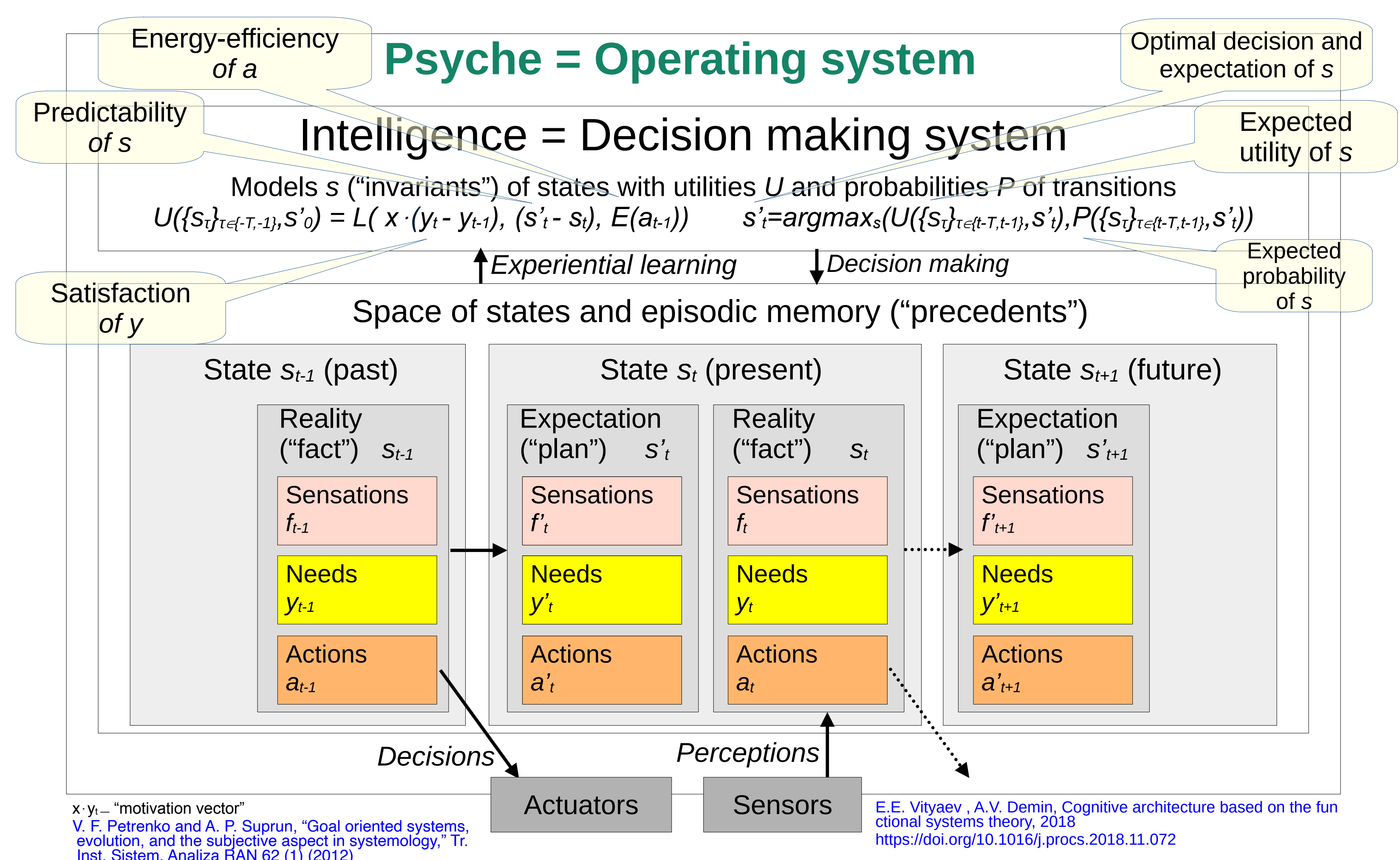
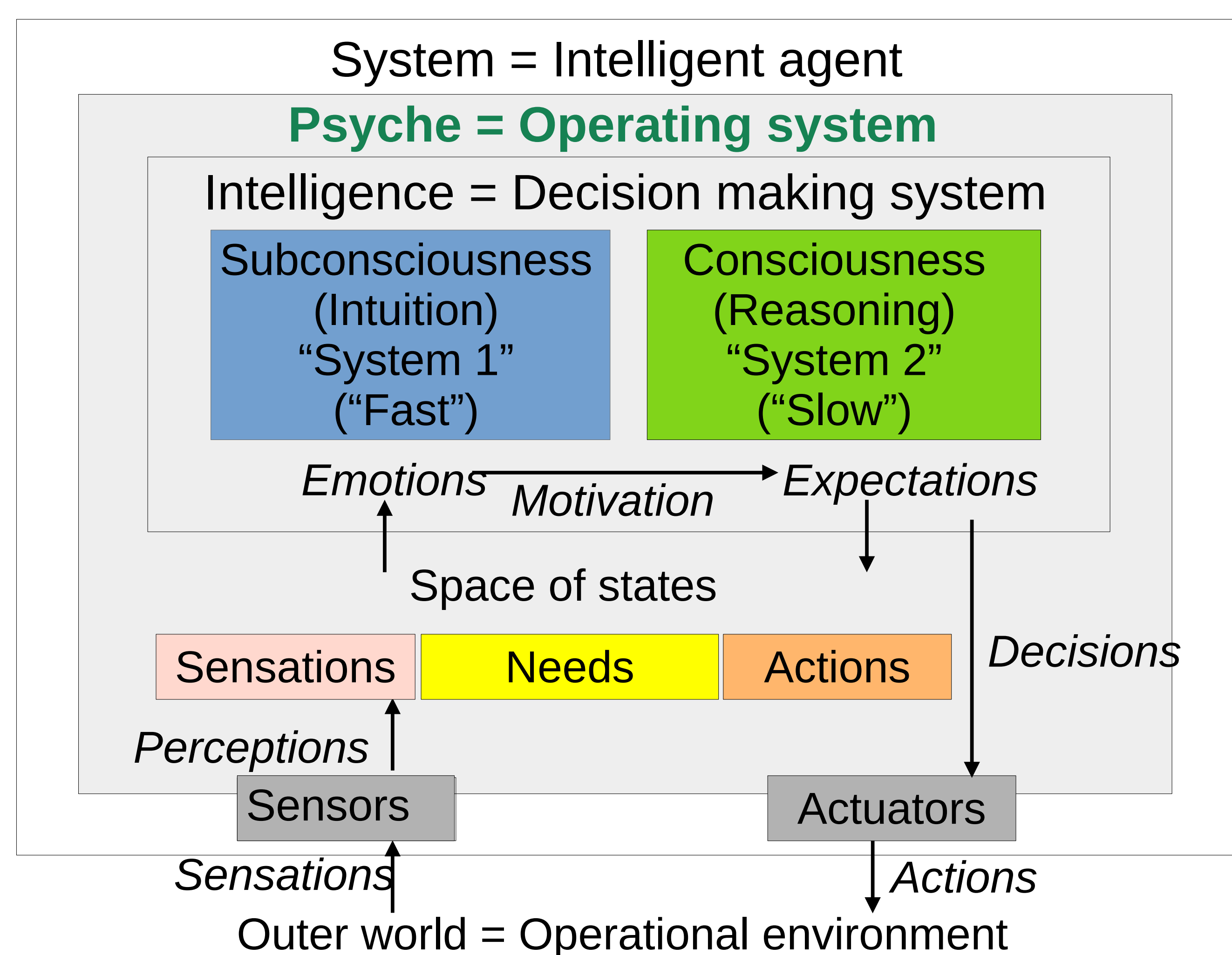
² Aigents, Pravdy 6, Novosibirsk, 630090, Russia, <https://aigents.com/>

³ SingularityNET Foundation, Baarerstrasse 141, Zug, 6300, Switzerland, <https://singularitynet.io>

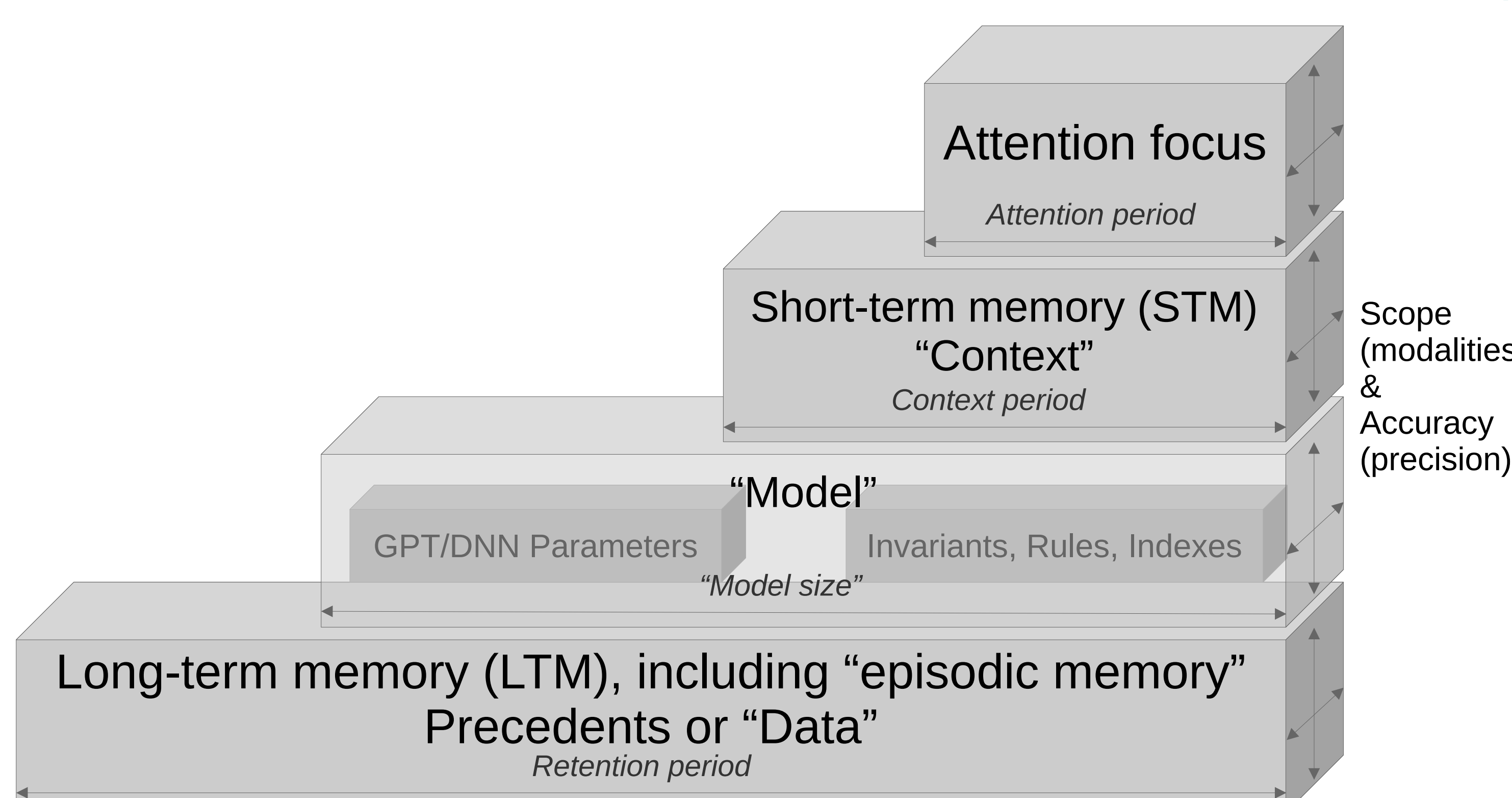
Abstract. This article presents an overview of approaches to modeling the human psyche in the context of constructing an artificial one. Based on this overview, a concept of cognitive architecture is proposed, in which the psyche is viewed as the operating system of a living or artificial subject, comprising a space of states, including the state of needs that determine the meaning of a subject's being in relation to stimuli from the external world, and intelligence as a decision-making system regarding actions in this world to satisfy these needs. Based on this concept, a computational formalization is proposed for creating artificial general intelligence systems for an agent through experiential learning in a state space that includes agent's needs, taking into account their biological or existential significance for the intelligent agent, along with agent's sensations and actions. Thus, the problem of constructing artificial general intelligence is formalized as a system for making optimal decisions in the space of specific agent needs under conditions of uncertainty, maximizing success in achieving goals, minimizing existential risks, and maximizing energy efficiency. A minimal experimental implementation of the model is presented.



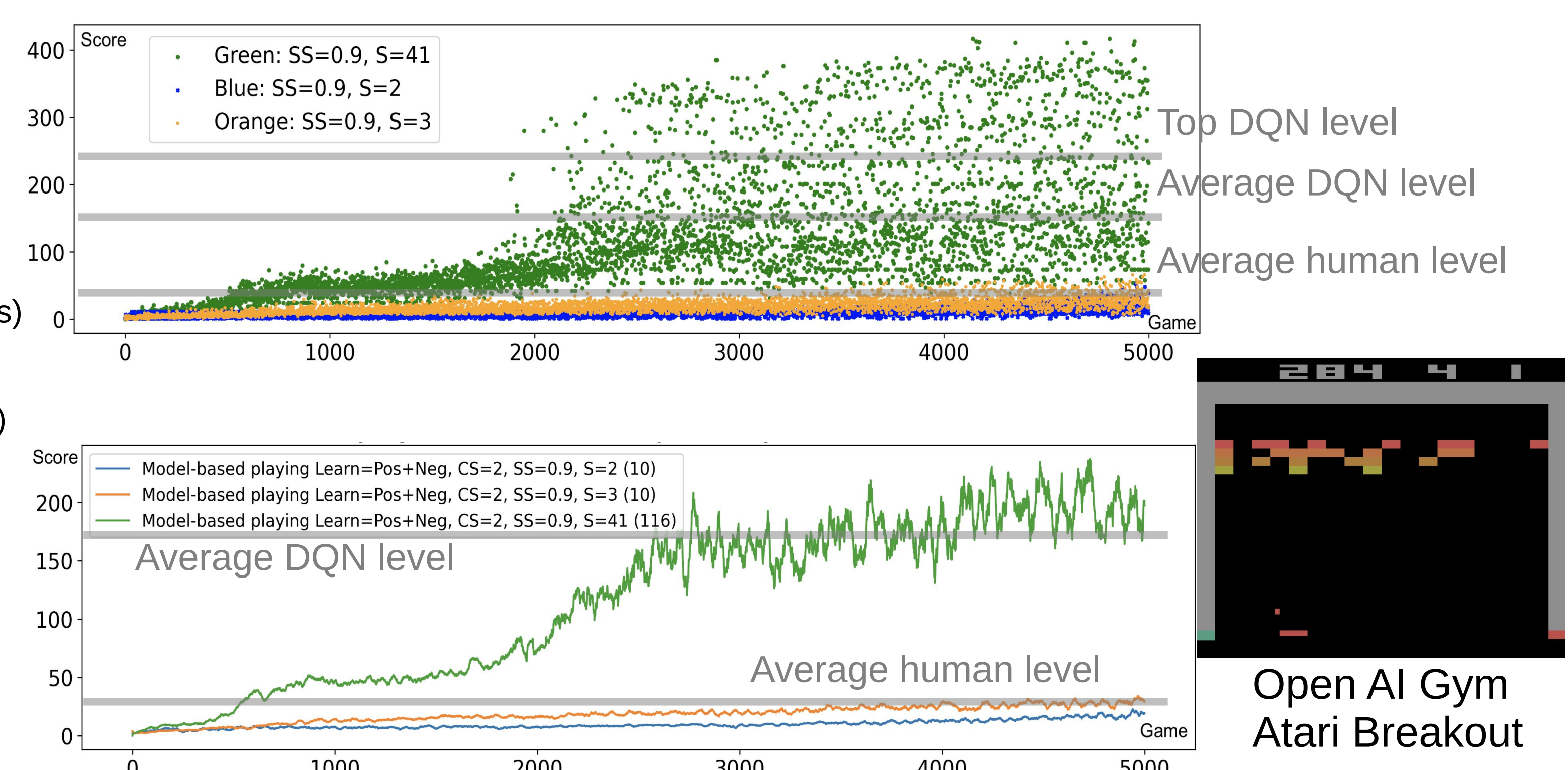
Keywords: artificial agent · artificial general intelligence · cognitive architecture · experiential learning · psyche model · space of needs · space of states.



Attention, context, model and memories



Reinforcement learning – experiential learning and decision making based and transitions between state series and “global feedback”



<https://arxiv.org/abs/2603.15586>

<https://arxiv.org/abs/2605.00940>